



MERN STACK INTERN

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Task Report: (Advanced CSS)

Project: Responsive Feature/Team Showcase

Module: Week 2, Day 1 – Advanced CSS

Focus: Transforms, Transitions, and Keyframe Animations

GitHub Link : <https://github.com/955muneeb/Glaxit-internship/tree/main/week-1>

1. Introduction

The first week of the MERN Stack Internship focused on strengthening the core concepts of frontend development.

Throughout the week, I explored HTML5 semantic elements, CSS fundamentals, responsive layouts using Grid and Flexbox, media queries, and mobile-first development. The week concluded with the deployment of an existing React portfolio, demonstrating how these foundational concepts integrate into modern component-based applications.

2. Objectives

The objectives of Week 1 were:

- Understand semantic HTML5.
 - Build accessible webpage structures.
 - Learn client-side form validation.
 - Master the CSS Box Model.
 - Create responsive layouts.
 - Learn CSS Grid.
 - Learn CSS Flexbox.
 - Build mobile-first webpages.
 - Apply responsive media queries.
 - Understand scalable CSS architecture.
 - Connect HTML and CSS knowledge with React applications.
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3. Day 1 – Semantic HTML & Forms

Semantic HTML

The project was developed using semantic HTML5 elements instead of relying entirely on generic `<div>` elements.

The following semantic tags were implemented:

- Header
- Navigation

- Main
- Section
- Article
- Footer

This structure closely follows industry standards for professional web development.

HTML Forms

A contact form was implemented using native HTML validation.

The form included:

- Name
- Email
- Phone Number
- Subject
- Message

Validation techniques included:

- Required fields
 - Minimum character limits
 - Phone number patterns
 - Email validation
 - Proper input types
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4. Day 2 – CSS Fundamentals & Box Model

The second day focused on applying professional styling practices using CSS.

Topics covered included:

- | | |
|-----------------|----------------------|
| • CSS selectors | • Margins |
| • Typography | • Padding |
| • Colors | • Borders |
| • Spacing | • Display properties |
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Typography

Google Fonts (Inter) was integrated to improve readability and maintain a modern appearance.

Additional styling included:

- Font hierarchy
- Line spacing
- Consistent font sizing
- Hover effects
- Focus effects
- Smooth transitions

These improvements enhance user experience and visual consistency.

5. Day 3 – CSS Grid & Flexbox

Modern websites require efficient layout systems. Therefore, both CSS Grid and Flexbox were studied and implemented.

CSS Grid

Grid was used for creating the overall webpage structure.

The page layout included:

- Header
- Navigation
- Main Content
- Sidebar
- Footer

Grid simplifies two-dimensional layouts and allows components to align consistently across the page.

6. Day 4 – Responsive Web Design

One of the major focuses of Week 1 was responsive web development.

The project adopted a **Mobile-First** approach.

Media Queries

Responsive layouts were implemented using:

- 768px breakpoint
- 1024px breakpoint

The interface progressively changed from:

- Single-column layout
- Two-column layout
- Three-column layout

This ensured compatibility across:

- Mobile phones
 - Tablets
 - Laptops
 - Desktop computers
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Relative Units

Pixels were replaced with rem units to improve scalability.

7. Day 5 – Mini Project (React Portfolio)

<https://muneebrehmanportfolio.netlify.app/>

Since I already possess practical experience with React.js and the MERN Stack, the final day was utilized to showcase an existing portfolio project deployed on Netlify instead of creating another basic static HTML project.

The deployed portfolio demonstrates practical knowledge of:

- React.js

- Component-based architecture
- Responsive layouts
- Modern UI development
- Routing
- Deployment workflow

This capstone activity effectively connects the frontend fundamentals learned throughout the week with real-world React development practices.

11. Conclusion

Overall, this week has prepared me for upcoming topics in the MERN Stack Internship, including advanced JavaScript, browser APIs, version control with Git, React development, backend technologies, and full-stack application development.